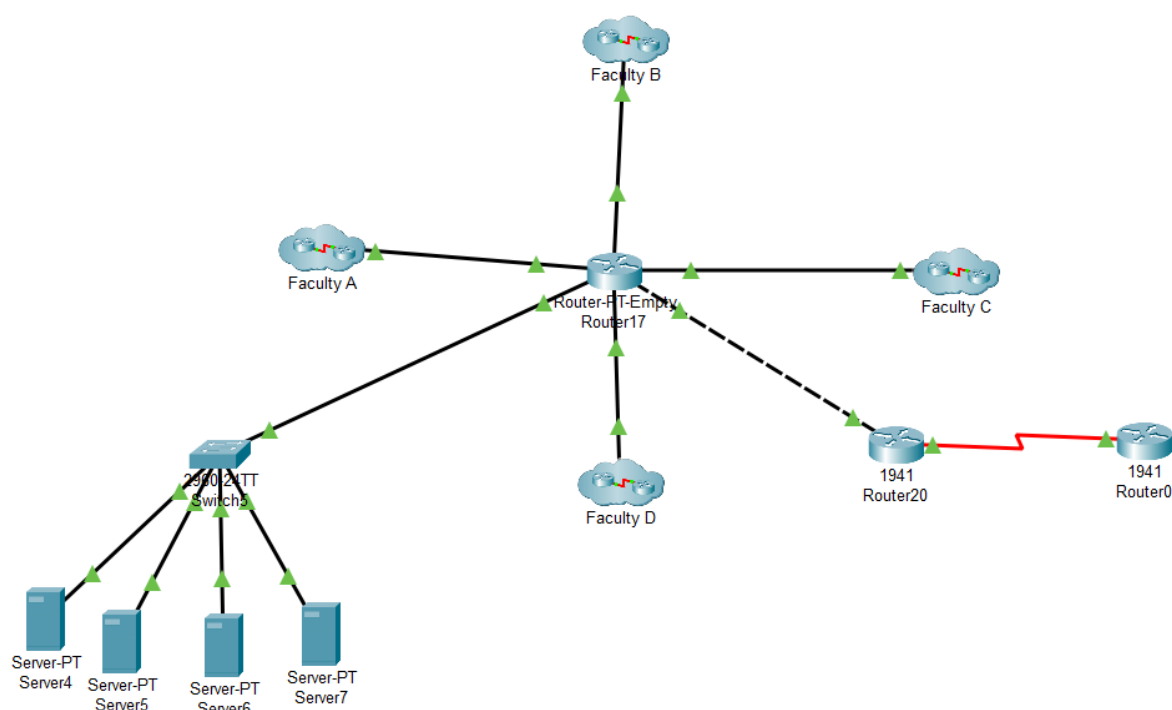


University Campus Network Design Report

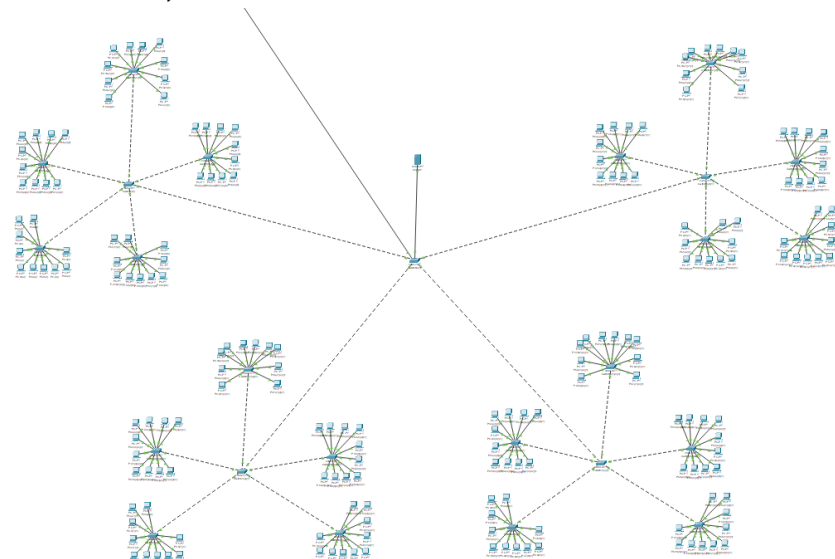
1. Introduction

- **Objective:** Design a network for a small university campus.
- **Purpose:** Connect multiple faculties, labs, and a Scientific Computation Center efficiently, ensuring access to internal and external networks.
- **Scope:** Includes routing, VLAN implementation, and star topologies.

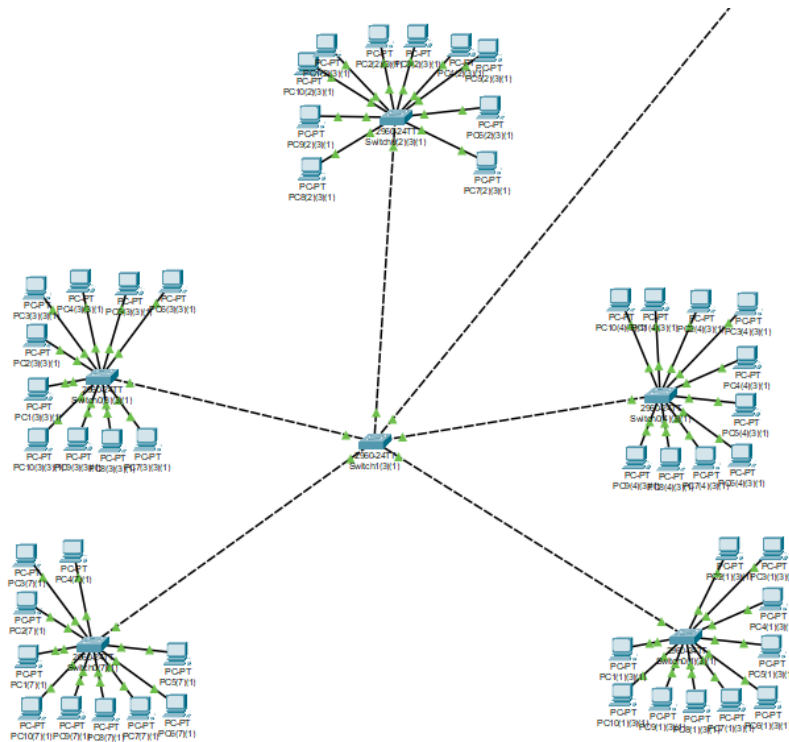


2. Network Design Requirements

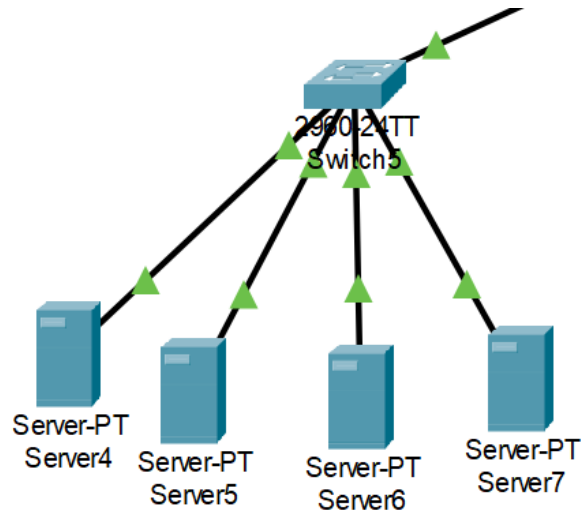
- **Faculties:** 4 faculties, each with 4 floors.



- **Labs:** Each floor contains 5 labs, each with 10 PCs connected via switches.



- **Scientific Computation Center:** Contains several servers connected by a switch.



- **Connections:**
 - All lab switches on a floor connect to a floor switch.
 - Floor switches connect to a main faculty switch.
 - The main faculty switch and Scientific Computation Center connect to the university router.

- **External Connectivity:** The university router links to the World Wide Web.

3. IP Addressing

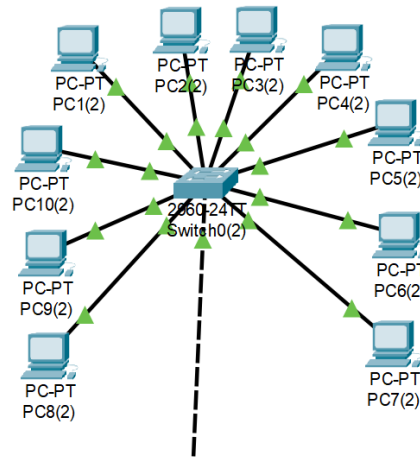
- Classless IP addressing optimizes the use of public IP addresses across networks.

Device Name: Router17				
Device Model: Router-PT-Empty				
Hostname: Router				
Port	Link	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	192.168.1.1/24	<not set>	00E0.8F93.8AC1
GigabitEthernet1/0	Up	192.168.2.1/24	<not set>	0002.4A5B.B7C6
GigabitEthernet2/0	Up	192.168.3.1/24	<not set>	0009.7C9E.E4B9
GigabitEthernet3/0	Up	192.168.4.1/24	<not set>	0005.5E1E.773C
GigabitEthernet4/0	Up	192.168.5.1/24	<not set>	0005.5ED7.24ED
GigabitEthernet5/0	Up	192.168.6.1/24	<not set>	0050.0FBC.15B5
GigabitEthernet6/0	Down	<not set>	<not set>	0003.E43D.9A13
GigabitEthernet7/0	Down	<not set>	<not set>	0001.9738.37BD
GigabitEthernet8/0	Down	<not set>	<not set>	0001.43A3.6E00
GigabitEthernet9/0	Down	<not set>	<not set>	000A.4118.423E

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router17

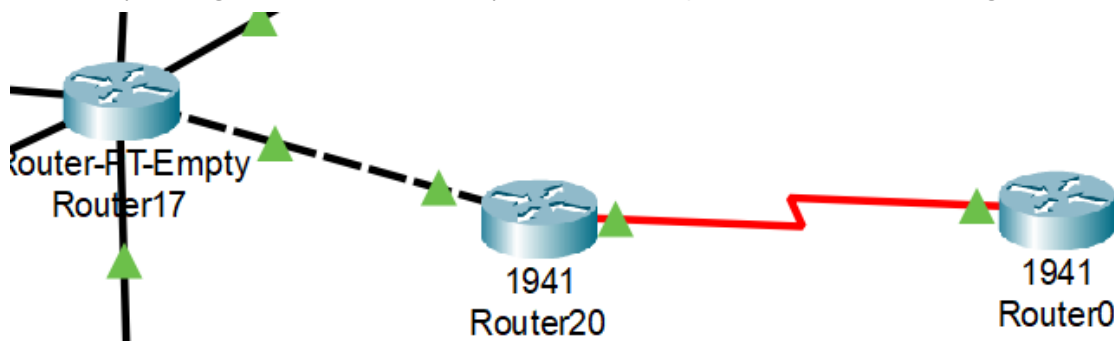
4. Topologies

- **Labs:** star topologies.

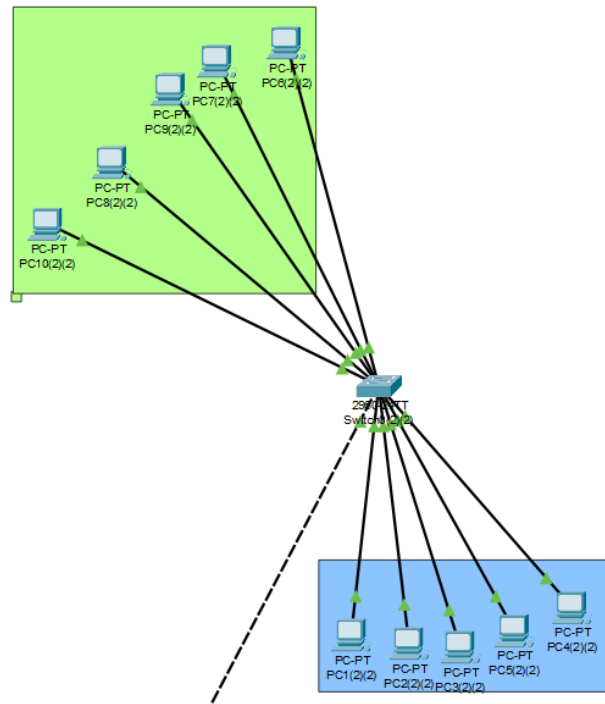


5. Routing and VLAN

- RIP (Routing Information Protocol) for inter-faculty and center data routing.



- VLAN configurations to segment network traffic efficiently.



6. Test Scenarios

- **Scenario 1:** A PC from Lab 2 on the first floor of the Faculty of Computers & AI accesses data from Server 0 in the Scientific Computation Center.
 - Source IP: PC 0 IP.
 - Destination IP: Server 0 IP.
- **Scenario 2:** VLAN communication is tested between labs of different faculties.
- **Scenario 3:** Internet connectivity is verified for a lab PC.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	FA lab1	Server4	ICMP		0.000	N	0	(edit)
	Successful	FA lab1	FB Lab1	ICMP		0.000	N	1	(edit)
	Successful	FA lab1	Server7	ICMP		0.000	N	2	(edit)